

1 Tools from Sheaf Theory

1.1 Presheaf and Sheaf

Definition 1. A **presheaf** $\mathcal{S}$ on a topological space X is an assignment to each open set $U \subseteq X$ of a set $\mathcal{S}(U)$ along with a map $r_V^U : \mathcal{S}(U) \rightarrow \mathcal{S}(V)$ whenever $V \subseteq U$ satisfying the following two properties:

- (a) $r_U^U = \text{Id}_U : \mathcal{S}(U) \rightarrow \mathcal{S}(U)$ for any open set $U \subseteq X$;
- (b) $r_W^V \circ r_V^U = r_W^U : \mathcal{S}(U) \rightarrow \mathcal{S}(W)$ whenever $W \subseteq V \subseteq U$ are the open sets in X .

$\mathcal{S}(U)$ is called the set of **sections of $\mathcal{S}$ over U** . A presheaf $\mathcal{S}$ on X is called a **presheaf of abelian groups** if each $\mathcal{S}(U)$ is endowed with an abelian group structure and the maps $r_V^U : \mathcal{S}(U) \rightarrow \mathcal{S}(V)$ are group homomorphisms.

Similarly, we can define **presheaf of rings** and **presheaf of vector spaces**.

In the rest of this note, a *presheaf* will always refer to presheaf of abelian groups if there is no special statement.

Notation: For a presheaf $\mathcal{S}$ on X and an open sets $V \subseteq U$ in X , we denote

$$s|_V := r_V^U(s) \quad \text{for } \forall s \in \mathcal{S}(U)$$

Remark: So if we use the language of category, let $\mathbf{Top}(X)$ be the category whose objects are open subsets of X and whose morphisms are set inclusions, let $\mathbf{C}$ be a category (usually, $\mathbf{C}$ would be sets, abelian groups, rings and so on), then a **presheaf on X with values in $\mathbf{C}$** is a contravariant functor from $\mathbf{Top}(X)$ to $\mathbf{C}$.

Definition 2. A **sheaf** on the topological space X is a presheaf $\mathcal{S}$ satisfying the following two properties: Let U be an open subset of X and $\{U_\alpha\}_{\alpha \in I}$ is a family of open subsets in X such that $U = \bigcup_{\alpha \in I} U_\alpha$

(**Locality Property**) if $s, t \in \mathcal{S}(U)$ such that $s|_{U_\alpha} = t|_{U_\alpha}$ ($\forall \alpha \in I$), then we have $s = t$;

(**Gluing Property**) Given some elements $s_\alpha \in \mathcal{S}(U_\alpha)$, if $s_\alpha|_{U_\alpha \cap U_\beta} = s_\beta|_{U_\alpha \cap U_\beta}$ whenever $U_\alpha \cap U_\beta \neq \emptyset$, then we can find an element $s \in \mathcal{S}(U)$ such that $s|_{U_\alpha} = s_\alpha$ ($\forall \alpha \in I$).

Definition 3. For a given sheaf $\mathcal{S}$ on X , let $\mathcal{T}$ be another sheaf on X . If for any open subset U of X , $\mathcal{T}(U)$ is a subgroup of $\mathcal{S}(U)$ and $r_V^U(\mathcal{T}(U)) \subseteq \mathcal{T}(V)$ whenever $V \subseteq U$, then $\mathcal{T}$ is called a **subpresheaf** of $\mathcal{S}$. Moreover, if $\mathcal{T}$ satisfies the locality and gluing properties, it is called a **subsheaf** of $\mathcal{S}$.

Here are some useful examples of sheaves, which will be frequently used.

Example 1. Let M be a smooth manifold, for any open subset $U \subseteq M$, define

$$\mathcal{C}(U) := \{f \mid f : U \rightarrow \mathbb{C} \text{ is continuous}\}$$

and r_V^U is defined to be the usual restriction maps. Then $\mathcal{C}$ is a sheaf on M .

Example 2. Let M be a smooth manifold, for any open subset $U \subseteq M$, define

$$\mathcal{C}^*(U) := \{f \mid f : U \rightarrow \mathbb{C} \setminus \{0\} \text{ is continuous}\}$$

and r_V^U is defined to the usual restriction maps. Then $\mathcal{C}^*(U)$ is an abelian group under pointwise multiplication and $\mathcal{C}^*$ is a sheaf on M .

Example 3. Let M be a smooth manifold, for any open subset $U \subseteq M$, define

$$\mathcal{E}(U) := \{f \mid f : U \rightarrow \mathbb{C} \text{ is smooth}\}$$

r_V^U is the usual restriction map. Then $\mathcal{E}$ is sheaf on M .

Example 4. Let M be a smooth manifold, for any open subset $U \subseteq M$, define

$$\mathcal{E}^k(U) := \Gamma(\wedge_{\mathbb{C}}^k(U)), \quad \mathcal{Z}^k(U) := \ker d|_{\mathcal{E}^k(U)}$$

and r_V^U of both is defined to be the usual restriction maps. Then $\mathcal{E}^k$ and $\mathcal{Z}^k$ are sheaves on M , moreover $\mathcal{Z}^k$ is a subsheaf of $\mathcal{E}^k$. r_V^U is the usual restriction map. Then $\mathcal{E}$ is a sheaf on M .

Example 5. Let M be a smooth manifold and $E \rightarrow M$ is a smooth complex vector bundle, for any open subset $U \subseteq M$, define

$$\mathcal{E}(E)(U) := \Gamma(U; E)$$

i.e., the space of smooth sections of E on U , r_V^U is the usual restriction map. Then $\mathcal{E}(E)$ is a sheaf on M .

Example 6. Let M be a smooth manifold and $E \rightarrow M$ is a smooth complex vector bundle, for any open subset $U \subseteq M$, define

$$\mathcal{E}^k(E)(U) := \mathcal{E}^k(U; E)$$

i.e., the space of smooth E -valued complex k -forms on U and r_V^U is defined to be the usual restriction maps. Then $\mathcal{E}^k(E)$ is a sheaf on M .

Remark: for the bundle-valued forms, see «Chern Connections» .

Example 7. Let M be a smooth manifold and let G be an abelian group, for any open subset $U \subseteq M$, define

$$\underline{G}(U) := \{f \mid f : U \rightarrow G \text{ is locally constant}\}$$

and r_V^U is defined to be the usual restriction maps. Then $\underline{G}(U)$ is an abelian group under pointwise addition and $\underline{G}$ is a sheaf on M , which is called **constant sheaf** with coefficients in G .

Example 8. Let M be a smooth manifold and let G be an abelian group, then for any open subset $U \subseteq M$, define

$$G_p(U) := \begin{cases} G & \text{if } p \in U \\ \{0\} & \text{if } p \notin U \end{cases}$$

and r_V^U is defined to be

$$r_V^U := \begin{cases} \text{Id}_G & \text{if } p \in V \subseteq U \\ 0 & \text{else} \end{cases}$$

such sheaf is called a **skyscraper sheaf**.

Example 9. Let M be a complex manifold, for any open subset $U \subseteq M$, define

$$\mathcal{O}(U) := \{f \mid f : U \rightarrow \mathbb{C} \text{ is holomorphic}\}$$

and r_V^U is defined to be the usual restriction map. Then $\mathcal{O}$ is a sheaf on M .

Example 10. Let M be a complex manifold, for any open subset $U \subseteq M$, define

$$\mathcal{O}^*(U) := \{f \mid f : U \rightarrow \mathbb{C} \setminus \{0\} \text{ is holomorphic}\}$$

and r_V^U is the usual restriction map. Then $\mathcal{O}^*(U)$ is an abelian group under pointwise multiplication and $\mathcal{O}^*$ is a sheaf on M .

Example 11. Let M be a complex manifold, for any open subset $U \subseteq M$, define

$$\mathcal{E}^{p,q}(U) := \Gamma(\wedge^{p,q}(U)), \quad \mathcal{Z}^{p,q}(U) := \ker \bar{\partial}|_{\mathcal{E}^{p,q}(U)}$$

and r_V^U of both is defined to be the usual restriction map. Then $\mathcal{E}^{p,q}$ and $\mathcal{Z}^{p,q}$ are sheaves on M .

Example 12. Let M be a complex manifold, for any open subset $U \subseteq M$, define

$$\Omega^p(U) := \mathcal{O}(\wedge^{p,0}(U))$$

i.e., $\Omega^p(U)$ is the space of holomorphic p -forms on U , r_V^U is the usual restriction map. Then Ω^p is a sheaf on M and hence Ω^p is the same sheaf as $\mathcal{Z}^{p,0}$.

Example 13. Let M be a complex manifold and $E \rightarrow M$ a holomorphic vector bundle, for any open subset $U \subseteq M$, define

$$\mathcal{O}(E)(U) := \mathcal{O}(U; E)$$

i.e., $\mathcal{O}(E)(U)$ is the space of holomorphic sections of E over U and r_V^U is the usual restriction map. Then $\mathcal{O}(E)$ is a sheaf on M .

Example 14. Let M be a complex manifold and $E \rightarrow M$ a holomorphic vector bundle, for any open subset $U \subseteq M$, define

$$\mathcal{E}^{p,q}(E)(U) := \mathcal{E}^{p,q}(U; E), \quad \mathcal{Z}^{p,q}(E)(U) := \ker \bar{\partial}_E|_{\mathcal{E}^{p,q}(U; E)}$$

i.e., the space of smooth E -valued (p, q) -forms on U , and r_V^U is the usual restriction map. Then $\mathcal{E}^{p,q}(E)$ and $\mathcal{Z}^{p,q}(E)$ are sheaves on M .

Example 15. Let M be a complex manifold and $E \rightarrow M$ a holomorphic vector bundle, for any open subset $U \subseteq M$, define

$$\Omega^p(E)(U) := \Omega^p(U; E) = \mathcal{O}(\wedge^{p,0}(U) \otimes E)$$

i.e., the space of holomorphic E -valued p -forms on U , r_V^U is the usual restriction map. Then $\Omega^p(E)$ is a sheaf on M .

Moreover, we have the following definition of sheaf of modules over a sheaf of rings.

Definition 4. Let $\mathcal{R}$ be a sheaf of commutative rings with identity element on the topological space X . Let $\mathcal{S}$ be another sheaf on X , if for any open subset U in X , $\mathcal{S}(U)$ is an $\mathcal{R}(U)$ -module and whenever $V \subseteq U \subseteq X$ are open subsets we have

$$(rs)|_V = (r|_V)(s|_V) \quad \text{for } \forall r \in \mathcal{R}(U), \forall s \in \mathcal{S}(U)$$

then $\mathcal{S}$ is called a **sheaf of $\mathcal{R}$ -modules**.

Here are some examples of sheaves of modules.

Example 16. Let M be a smooth manifold and $E \rightarrow M$ a smooth complex vector bundle, then the $\mathcal{E}^k, \mathcal{E}(E), \mathcal{E}^k(E)$ are sheaves of $\mathcal{E}$ -modules.

Example 17. Let M be a complex manifold, $E \rightarrow M$ a holomorphic vector bundle, then $\mathcal{E}^{p,q}, \mathcal{E}^{p,q}(E)$ are sheaves of $\mathcal{E}$ -modules on M and $\Omega^p, \mathcal{O}(E)$ are sheaves of $\mathcal{O}$ -modules.

Example 18. Consider the constant sheaf $\mathbb{Z}$ on the topological space X . Let $\mathcal{S}$ be a sheaf on X , suppose $s \in \mathcal{S}(U)$ and $k \in \mathbb{Z}(U)$, then for any $x \in U$, there is a neighborhood $V_x \subseteq U$ of x such that $k|_{V_x}$ is a constant integer and therefore $(k|_{V_x})(s|_{V_x}) \in \mathcal{S}(V_x)$. Since $U = \bigcup_{x \in U} V_x$, so by the gluing property, there is a section, namely $ks \in \mathcal{S}(U)$ such that $(ks)|_{V_x} = (k|_{V_x})(s|_{V_x})$. So we see that every sheaf $\mathcal{S}$ of abelian groups on X is a sheaf of $\mathbb{Z}$ -modules.

1.2 Sheaf Morphisms

Definition 5. Let $\mathcal{S}, \mathcal{T}$ be two sheaves on the topological space X , then a **sheaf morphism** from $\mathcal{S}$ to $\mathcal{T}$

$$F : \mathcal{S} \longrightarrow \mathcal{T}$$

is defined to be a collection of group homomorphisms

$$F_U : \mathcal{S}_U \longrightarrow \mathcal{T}(U) \quad \text{for every open subset } U \text{ in } X$$

which makes the following diagram commutes

$$\begin{array}{ccc} \mathcal{S}(U) & \xrightarrow{F_U} & \mathcal{T}(U) \\ r_V^U \downarrow & & \downarrow r_V^U \\ \mathcal{S}(V) & \xrightarrow{F_V} & \mathcal{T}(V) \end{array}$$

whenever we have the open subsets $V \subseteq U$ in X . Similarly, we can define sheaf morphisms in the category of vector spaces, rings, modules and so on.

Remark: If $\mathcal{S}$ and $\mathcal{T}$ are presheaves, then the **presheaf morphism** from $\mathcal{S}$ to $\mathcal{T}$ is defined precisely the same way as above.

Convention: If there is no confusion, we will denote the homomorphism F_U just by F . So in this convention, the commutative diagram reads

$$F(s|_V) = F(s)|_V \quad \text{for any } s \in \mathcal{S}(U)$$

Definition 6. For two sheaf morphisms $F : \mathcal{S} \rightarrow \mathcal{T}$ and $G : \mathcal{T} \rightarrow \mathcal{R}$, their composition

$$G \circ F : \mathcal{S} \longrightarrow \mathcal{R}$$

is given by

$$(G \circ F)_U := G_U \circ F_U : \mathcal{S}(U) \rightarrow \mathcal{R}(U)$$

Definition 7. A sheaf morphism $F : \mathcal{S} \rightarrow \mathcal{T}$ is called a **sheaf isomorphism** if there is a sheaf morphism $G : \mathcal{T} \rightarrow \mathcal{S}$ such that

$$F \circ G = \text{Id}_{\mathcal{T}}, \quad G \circ F = \text{Id}_{\mathcal{S}}$$

in which case we denote $\mathcal{S} \cong \mathcal{T}$.

Here are some examples of sheaf morphisms.

Example 19. Let $\mathcal{S}$ be a sheaf on the topological space X and let $\mathcal{T}$ be a subsheaf of $\mathcal{S}$. Then we have the **inclusion morphism**

$$i : \mathcal{T} \hookrightarrow \mathcal{S}$$

which is given by

$$i_U(\sigma) := \sigma$$

for all $\sigma \in \mathcal{T}(U)$.

Example 20. Let M be a smooth manifold, then for any $k \in \mathbb{N}$, then the exterior derivative operator

$$d : \mathcal{E}^k(M) \rightarrow \mathcal{E}^{k+1}(M)$$

acts locally on every open subset $U \subseteq M$

$$d_U : \mathcal{E}^k(U) \rightarrow \mathcal{E}^{k+1}(U)$$

it's clear that we have the commutative diagram

$$\begin{array}{ccc} \mathcal{E}^k(U) & \xrightarrow{d_U} & \mathcal{E}^{k+1}(U) \\ r_V^U \downarrow & & \downarrow r_V^U \\ \mathcal{E}^k(V) & \xrightarrow{d_V} & \mathcal{E}^{k+1}(V) \end{array}$$

where the maps r_V^U are the usual restriction maps. Therefore, we get a sheaf morphism

$$d : \mathcal{E}^k \longrightarrow \mathcal{E}^{k+1}$$

Example 21. Let M be a complex manifold, by a similar discussion like the example above, we know that there is a sheaf morphism

$$\bar{\partial} : \mathcal{E}^{p,q} \longrightarrow \mathcal{E}^{p,q+1}$$

like the example above.

Example 22. Let M be a complex manifold, then we have the sheaves $\mathcal{O}$ and $\mathcal{O}^*$. Then we can define a sheaf morphism $\varepsilon : \mathcal{O} \rightarrow \mathcal{O}^*$ as follows

$$\varepsilon_U : \mathcal{O}(U) \longrightarrow \mathcal{O}^*(U)$$

$$f \mapsto e^{2\pi\sqrt{-1}f}$$

hence

$$\varepsilon_U(f + g) = \varepsilon_U(f)\varepsilon_U(g)$$

Example 23. Let G, H be two abelian groups and then we have two constant sheaves $\underline{G}, \underline{H}$ on the topological space X . Suppose $F : G \rightarrow H$ is a group homomorphism, then it induces a sheaf morphism $\underline{F} : \underline{G} \rightarrow \underline{H}$ by defining

$$\underline{F}_U : \underline{G}(U) \longrightarrow \underline{H}(U)$$

$$f \mapsto F \circ f$$

for each open subsets $U \subseteq X$.

Remark: Let X be a topological space, then the sheaves on X together with the sheaf morphisms form a category $\mathbf{Sh}(X)$.

1.3 Stalks

Before we define the *stalks* associated to a presheaf or sheaf, we make some constructions in the algebraic sense.

Definition 8. Let I be a nonempty set with a relation $\leq$ that satisfies the following properties:

- (a) $\alpha \leq \alpha$ for $\forall \alpha \in I$;
- (b) $\alpha \leq \beta, \beta \leq \gamma \Rightarrow \alpha \leq \gamma$ for $\alpha, \beta, \gamma \in I$;
- (c) for any $\alpha, \beta \in I$, there exists $\gamma \in I$ such that $\alpha \leq \gamma$ and $\beta \leq \gamma$. such $(I, \leq)$ is called **directed set**.

Definition 9. A **direct system** of algebraic objects (usually, it would be groups, rings, modules and so on) is a family $\{G_\alpha\}_{\alpha \in I}$ of algebraic objects indexed by a directed set $(I, \leq)$ together with homomorphisms

$$f_{\alpha\beta} : G_\alpha \longrightarrow G_\beta$$

whenever $\alpha \leq \beta$, which satisfies

$$f_{\alpha\alpha} = \text{Id}_{G_\alpha}, \quad f_{\beta\gamma} \circ f_{\alpha\beta} = f_{\alpha\gamma} \quad \text{whenever } \alpha \leq \beta \leq \gamma$$

Now we make an equivalence relation $\sim$ on $\bigsqcup_{\alpha \in I} G_\alpha$ as follows:

$$g_\alpha (\in G_\alpha) \sim g_\beta (\in G_\beta) \iff \exists \gamma \in I \text{ such that } \alpha \leq \gamma, \beta \leq \gamma \text{ and } f_{\alpha\gamma}(g_\alpha) = f_{\beta\gamma}(g_\beta) \in G_\gamma$$

then we have the definition of direct limit.

Definition 10. Let $\{G_\alpha\}_{\alpha \in I}$ be a directed system and the equivalence relation $\sim$ is defined as above, then we define the **direct limit** to be

$$\lim_{\rightarrow} G_\alpha := \bigsqcup_{\alpha \in I} G_\alpha / \sim$$

the addition on $\lim_{\rightarrow} G_\alpha$ is given by

$$[g_\alpha] + [g_\beta] := [f_{\alpha\gamma}(g_\alpha) + f_{\beta\gamma}(g_\beta)]$$

where α, β, γ is as above.

Apply these algebraic constructions to sheaves.

Definition 11. Let X be a topological space and $\mathcal{S}$ a sheaf on X . Let $\{U\}$ be the open subsets in X , then it is a directed set under the relation defined by

$$U \leq V \iff U \supseteq V$$

then $\{\mathcal{S}(U)\}$ becomes a direct system with the maps r_V^U . Now we define the **stalk of $\mathcal{S}$ at p** to be

$$\mathcal{S}_p := \lim_{\substack{\rightarrow \\ p \in U}} \mathcal{S}(U)$$

the equivalence class for $s \in \mathcal{S}(U)$ in $\mathcal{S}_p$ is denoted by $[s]_p$.

Remark: Let $\mathcal{S}, \mathcal{T}$ be two presheaves on X , then for any $p \in X$, the presheaf morphism $F : \mathcal{S} \rightarrow \mathcal{T}$ induces a **stalk homomorphism**

$$\begin{aligned} F_p : \mathcal{S}_p &\longrightarrow \mathcal{T}_p \\ [s]_p &\mapsto [F(s)]_p \end{aligned}$$

(note that we write $F(s)$ for $F_U(s)$ if $s \in \mathcal{S}(U)$ and if no confusion arise) it's easy to check that F_p is well-defined: let $s \in \mathcal{S}(U), t \in \mathcal{S}(V)$ such that $[s]_p = [t]_p$, then there exists $p \in W \subseteq U \cap V$ such that $s|_W = t|_W$, hence by the commutativity of the sheaf morphism and the restriction maps, we have $F(s)|_W = F(s|_W) = F(t|_W) = F(t)|_W$, which means that $[F(s)]_p = [F(t)]_p$.

1.4 Étale Space and Sheafification

By using the concept of stalks defined above, we can construct a topological space associated to any presheaf, which actually provide us a mechanism for turning a presheaf into sheaf.

Definition 12. Let X be a topological space, an **étalé space** of abelian groups over X is a topological space E together with a local homeomorphism $\pi : E \rightarrow X$ such that they satisfy the following properties:

- (1) for any $p \in M$, the preimage $E_p := \pi^{-1}(p)$ is an abelian group;
- (2) the negation $E \rightarrow E, s \mapsto -s$ and the group addition $E \times_X E \rightarrow E, (s_1, s_2) \mapsto s_1 + s_2$ are both continuous map, where $E \times_X E := \{(x_1, x_2) \in E \times E \mid \pi(x_1) = \pi(x_2)\} \subseteq E \times E$ is given the subspace topology.

Similarly, we can define the étalé space of rings, vector spaces and so on.

Let $\mathcal{S}$ be a presheaf on the topological space X , now we denote

$$\text{Et}(\mathcal{S}) := \bigsqcup_{p \in X} \mathcal{S}_p$$

and we denote the projection map

$$\begin{aligned} \pi : \text{Et}(\mathcal{S}) &\longrightarrow X \\ [s]_p &\mapsto p \end{aligned}$$

now, for each open subset U in X and each $s \in \mathcal{S}(U)$, we define a map

$$\begin{aligned} s^+ : U &\longrightarrow \text{Et}(\mathcal{S}) \\ p &\mapsto [s]_p \end{aligned}$$

then we have the following important result.

Theorem 1. Let $\mathcal{S}$ be a presheaf on the topological space X , then $\text{Et}(\mathcal{S})$ can be given a unique topology, which turns $\text{Et}(\mathcal{S})$ together with the projection map $\pi : \text{Et}(\mathcal{S}) \rightarrow X$ into an étalé space over X and the map $s^+ : U \rightarrow \text{Et}(\mathcal{S})$ defined above is continuous.

Proof: **(Sketch)** The idea is to make the collection of all subsets of the form

$$s^+(U) = \{[s]_p \mid p \in U\}$$

where U ranges over all open subsets in X and s ranges over all sections $s \in \mathcal{S}(U)$ as a basis for a topology on $\text{Et}(\mathcal{S})$. So in order to prove this, by the knowledge of general topology, we just need to prove the following two things:

(a) For any point $[s]_p \in \mathcal{S}_p \subseteq \text{Et}(\mathcal{S})$, it's clear that $[s]_p$ is in some $s^+(U)$;

(b) Let $[w]_p \in s^+(U) \cap t^+(V)$, so we know $p \in U \cap V$ and $[w]_p = [s]_p = [t]_p$, hence $\exists W \subseteq U \cap V$ such that $w|_W = s|_W = t|_W$ and hence we get $[w]_p \in w^+(W) \subseteq s^+(U) \cap t^+(V)$.

So by (a) and (b), we know that $\{s^+(U) \mid U \text{ open in } X \text{ and } s \in \mathcal{S}(U)\}$ is a basis for a topology on $\text{Et}(\mathcal{S})$.

Now we prove that $s^+ : U \rightarrow \text{Et}(\mathcal{S})$ is continuous: take any basis open subset $t^+(V)$ in $\text{Et}(\mathcal{S})$, since

$$(s^+)^{-1}(t^+(V)) = \{p \in U \cap V \mid [s]_p = [t]_p\} = \{p \in U \cap V \mid s|_W = t|_W \text{ for some neighborhood } W \subseteq U \cap V\}$$

which is open in U , so the map s^+ is continuous.

Then we prove that the projection map $\pi : \text{Et}(\mathcal{S}) \rightarrow X$ is a local homeomorphism: take any basis open subset $s^+(U)$ and let $V \subseteq U$ be open, since

$$\pi^{-1}(V) \cap s^+(U) = \{[s]_q \mid q \in V\} = (s|_V)^+(V)$$

which is exactly a basis open subset, hence we know that

$$\pi|_{s^+(U)} : s^+(U) \rightarrow U$$

is continuous and clearly that its inverse is given by the map $s^+ : U \rightarrow s^+(U)$, which is continuous and hence $\pi : \text{Et}(\mathcal{S}) \rightarrow X$ is a local homeomorphism.

We prove that the topology is unique: Suppose $\text{Et}(\mathcal{S})$ is endowed with another topology that makes π a local homeomorphism and s^+ continuous, in which we denote the topological space as $\widehat{E}$. Take $\forall \xi \in s^+(U)$ and assume $\pi(\xi) = p$, i.e., $\xi \in \mathcal{S}_p$. Since π is a local homeomorphism, let $\widehat{W}$ be an open neighborhood of ξ such that $\pi|_{\widehat{W}}$ is a homeomorphism and denote $W := \pi(\widehat{W})$.

1.5 Exact Sequence of Sheaves

1.6 Sheaf Cohomology

Definition 13. Let X be a topological space and $\mathcal{S}$ a sheaf of abelian groups on X . Suppose $\mathcal{U} = \{U_\alpha\}_{\alpha \in I}$ is an indexed open cover of X , for any $p \in \mathbb{Z}_+$, a **p -cochain on $\mathcal{U}$ with coefficients in $\mathcal{S}$** (or: **Cech cochain**) is defined to be an operator

$$c : (\alpha_0, \alpha_1, \dots, \alpha_p) \mapsto c_{\alpha_0 \alpha_1 \dots \alpha_p} \in \mathcal{S}(U_{\alpha_0} \cap U_{\alpha_1} \cdots \cap U_{\alpha_p})$$

where $(\alpha_0, \alpha_1, \dots, \alpha_p)$ is any multi-index of length $p + 1$.

Definition 14. The set of all such p -cochains is denoted by $C^p(\mathcal{U}; \mathcal{S})$, which is called the **p -th cochain group on $\mathcal{U}$ with coefficients in $\mathcal{S}$** . And it is an abelian group under the addition operation defined by

$$(c + c')_{\alpha_0 \dots \alpha_p} := c_{\alpha_0 \dots \alpha_p} + c'_{\alpha_0 \dots \alpha_p}$$

when $p < 0$, $C^p(\mathcal{U}; \mathcal{S}) := \{0\}$

Definition 15. We define the **coboundary operator** to be the homomorphism as follows

$$\begin{aligned}\delta : C^p(\mathcal{U}; \mathcal{S}) &\longrightarrow C^{p+1}(\mathcal{U}; \mathcal{S}) \\ c &\mapsto \delta c\end{aligned}$$

where for any multi-index $(\alpha_0, \alpha_1, \dots, \alpha_{p+1})$, δc is given by

$$(\delta c)_{\alpha_0 \dots \alpha_{p+1}} := \sum_{j=0}^{p+1} (-1)^j c_{\alpha_0 \dots \widehat{\alpha}_j \dots \alpha_p} \big|_{U_{\alpha_0} \cap \dots \cap U_{\alpha_{p+1}}}$$

Lemma 1. For the coboundary operator δ defined above, we have $\delta \circ \delta = 0$.

Proof: Straightforward computation by the definition of δ . $\square$

Remark: So this lemma tells us that the cochain groups together with the coboundary operator form a complex:

$$0 \longrightarrow C^0(\mathcal{U}; \mathcal{S}) \xrightarrow{\delta} C^1(\mathcal{U}; \mathcal{S}) \xrightarrow{\delta} \dots \xrightarrow{\delta} C^p(\mathcal{U}; \mathcal{S}) \xrightarrow{\delta} C^{p+1}(\mathcal{U}; \mathcal{S}) \xrightarrow{\delta} \dots$$

and hence we have the following definition

Definition 16. The **Cech cohomology groups on $\mathcal{U}$ with coefficients in $\mathcal{S}$** is defined by

$$H^p(\mathcal{U}; \mathcal{S}) := \frac{\ker(\delta : C^p(\mathcal{U}; \mathcal{S}) \rightarrow C^{p+1}(\mathcal{U}; \mathcal{S}))}{\operatorname{Im}(C^{p-1}(\mathcal{U}; \mathcal{S}) \rightarrow C^p(\mathcal{U}; \mathcal{S}))}$$

Remark: A cochain $c \in C^p(\mathcal{U}; \mathcal{S})$ is called a **(Cech) cocycle** if $\delta c = 0$ and called a **(Cech) coboundary** if there is some $b \in C^{p-1}(\mathcal{U}; \mathcal{S})$ such that $c = \delta b$. The groups of all cocycles is denoted by $Z^p(\mathcal{U}; \mathcal{S})$ and the groups of all coboundaries is denoted by $B^p(\mathcal{U}; \mathcal{S})$.

Example 24. Let $\mathcal{S}$ be a sheaf on the topological space X and $\mathcal{U} := \{U_\alpha\}_{\alpha \in I}$ is an indexed open cover for X . Suppose $c \in C^0(\mathcal{U}; \mathcal{S})$, then by the definition of δ , we get

$$(\delta c)_{\alpha\beta} = c_\beta|_{U_\alpha \cap U_\beta} - c_\alpha|_{U_\alpha \cap U_\beta}$$

note that $H^0(\mathcal{U}; \mathcal{S}) = Z^0(\mathcal{U}; \mathcal{S})$ since $B^0(\mathcal{U}; \mathcal{S}) = \{0\}$. Consider a global section $\sigma \in \mathcal{S}(X)$, if we define

$$c_\alpha := \sigma|_{U_\alpha} \text{ for } \forall \alpha \in I$$

then we get a 0-cochain $c \in C^0(\mathcal{U}; \mathcal{S})$. By the property of restriction maps, for any multi-index (α, β) , we know that $(\delta c)_{\alpha\beta} = c_\beta|_{U_\alpha \cap U_\beta} - c_\alpha|_{U_\alpha \cap U_\beta} = \sigma|_{U_\alpha \cap U_\beta} - \sigma|_{U_\alpha \cap U_\beta} = 0$, which means that $\delta c = 0$ and hence $c \in Z^0(\mathcal{U}; \mathcal{S}) = H^0(\mathcal{U}; \mathcal{S})$. So, by this way, we get a map

$$I_{\mathcal{U}; \mathcal{S}} : \mathcal{S}(X) \longrightarrow H^0(\mathcal{U}; \mathcal{S}) = Z^0(\mathcal{U}; \mathcal{S})$$

$$\sigma \mapsto c$$

where $c_\alpha := \sigma|_{U_\alpha}$ as above.

Proposition 1. The map

$$I_{\mathcal{U}; \mathcal{S}} : \mathcal{S}(X) \longrightarrow H^0(\mathcal{U}; \mathcal{S})$$

defined in the example above is an isomorphism.

Proof: Since $\mathcal{S}$ is a sheaf on X , the injectivity follows from the locality property and the surjectivity follows from the gluing property. $\square$

Example 25. Let $\mathcal{S}$ be a sheaf on the topological space X and $\mathcal{U} := \{U_\alpha\}_{\alpha \in I}$ is an indexed open cover for X . Suppose $c \in C^1(\mathcal{U}; \mathcal{S})$, then by the definition of δ , we get

$$(\delta c)_{\alpha\beta\gamma} = c_{\beta\gamma}|_{U_\alpha \cap U_\beta \cap U_\gamma} - c_{\alpha\gamma}|_{U_\alpha \cap U_\beta \cap U_\gamma} + c_{\alpha\beta}|_{U_\alpha \cap U_\beta \cap U_\gamma}$$

or for short we simply write

$$(\delta c)_{\alpha\beta\gamma} = c_{\beta\gamma} - c_{\alpha\gamma} + c_{\alpha\beta} \text{ on } U_\alpha \cap U_\beta \cap U_\gamma$$

which implies that

$$c \in Z^1(\mathcal{U}; \mathcal{S}) \iff c_{\alpha\gamma} = c_{\alpha\beta} + c_{\beta\gamma} \text{ on } U_\alpha \cap U_\beta \cap U_\gamma \text{ for any } \alpha, \beta, \gamma$$

and

$$[c] = [c'] \in H^1(\mathcal{U}; \mathcal{S}) \iff \exists b \in C^0(\mathcal{U}; \mathcal{S}) \text{ such that } c_{\alpha\beta} - c'_{\alpha\beta} = b_\beta - b_\alpha \text{ on } U_\alpha \cap U_\beta \text{ for any } \alpha, \beta$$

Remark: Also, note that if $\mathcal{S}$ is a sheaf of abelian groups under multiplication, then the discussion

in the example above is expressed as

$$c \in C^1(\mathcal{U}; \mathcal{S}) \Rightarrow (\delta c)_{\alpha\beta\gamma} = c_{\beta\gamma} c_{\alpha\gamma}^{-1} c_{\alpha\beta} \text{ on } U_\alpha \cap U_\beta \cap U_\gamma$$

$$c \in Z^1(\mathcal{U}; \mathcal{S}) \iff c_{\alpha\gamma} = c_{\alpha\beta} c_{\beta\gamma} \text{ on } U_\alpha \cap U_\beta \cap U_\gamma$$

$$[c] = [c'] \in H^1(\mathcal{U}; \mathcal{S}) \iff \exists b \in C^0(\mathcal{U}; \mathcal{S}) \text{ such that } c_{\alpha\beta} = c'_{\alpha\beta} b_\beta b_\alpha^{-1} \text{ on } U_\alpha \cap U_\beta$$

these in fact are closely related to the line bundles in terms of their transition functions.

Example 26. Let M be a smooth manifold and $\omega \in \mathcal{Z}^k(M)$. Then by the ordinary Poincare lemma, we know that ω is locally exact and hence we can find an open cover $\mathcal{U} = \{U_\alpha\}_{\alpha \in I}$ and $\eta_\alpha \in \mathcal{E}^{k-1}(U_\alpha)$ for any $\alpha \in I$ such that $\omega|_{U_\alpha} = d(\eta_\alpha)$. If we define

$$\gamma_{\alpha\beta} := \eta_\beta|_{U_\alpha \cap U_\beta} - \eta_\alpha|_{U_\alpha \cap U_\beta}$$

then we get a 1-cochain $\gamma \in C^1(\mathcal{U}; \mathcal{Z}^{k-1})$ since $d\gamma_{\alpha\beta} = d(\eta_\beta|_{U_\alpha \cap U_\beta}) - d(\eta_\alpha|_{U_\alpha \cap U_\beta}) = (d\eta_\beta)|_{U_\alpha \cap U_\beta} - (d\eta_\alpha)|_{U_\alpha \cap U_\beta} = 0$. Therefore by the definition of δ , we get $\delta\gamma = 0$ by an easy computation, which means that $\gamma \in Z^1(\mathcal{U}; \mathcal{Z}^{k-1})$. Moreover, we can know that

$$\gamma \in B^1(\mathcal{U}; \mathcal{Z}^{k-1}) \iff \exists \sigma \in C^0(\mathcal{U}; \mathcal{Z}^{k-1}) \text{ such that } \gamma_{\alpha\beta} = \sigma_\beta|_{U_\alpha \cap U_\beta} - \sigma_\alpha|_{U_\alpha \cap U_\beta}$$

i.e.,

$$\gamma \in B^1(\mathcal{U}; \mathcal{Z}^{k-1}) \iff \exists \sigma \in C^0(\mathcal{U}; \mathcal{Z}^{k-1}) \text{ such that } \sigma_\beta - \sigma_\alpha = \eta_\beta - \eta_\alpha \text{ on } U_\alpha \cap U_\beta$$

so if $\gamma \in B^1(\mathcal{U}; \mathcal{Z}^{k-1})$, then $(\eta_\alpha - \sigma_\alpha)|_{U_\alpha \cap U_\beta} = (\eta_\beta - \sigma_\beta)|_{U_\alpha \cap U_\beta}$ and hence it piece together to give us a global $(k-1)$ -form, which we denote by θ , i.e., $\theta|_{U_\alpha} = \eta_\alpha - \sigma_\alpha$ and in this case, we get $(d\theta)|_{U_\alpha} = d(\theta|_{U_\alpha}) = d(\eta_\alpha - \sigma_\alpha) = \omega|_{U_\alpha} - 0 = \omega|_{U_\alpha}$ (here note that $\sigma_\alpha \in \mathcal{Z}^{k-1}(U_\alpha)$). Conversely, if ω is exact, i.e., $\exists \eta \in \mathcal{E}^{k-1}(M)$ such that $\omega = d\eta$, then we just define $\eta_\alpha := \eta|_{U_\alpha}$ for all $\alpha \in I$, which yields

$$\gamma_{\alpha\beta} = \eta_\beta|_{U_\alpha \cap U_\beta} - \eta_\alpha|_{U_\alpha \cap U_\beta} = 0 \text{ for any } \alpha, \beta$$

and thus $\gamma = 0$.

Remark: By this example, we know that for a given closed k -form $\omega \in \mathcal{Z}^k(M)$, it induces a 1-cocycle $\gamma \in Z^1(\mathcal{U}; \mathcal{Z}^{k-1})$ as above. Moreover, $\gamma \in B^1(\mathcal{U}; \mathcal{Z}^{k-1}) \iff \omega$ is exact. In fact, this is a special case of the connection between deRham cohomology and sheaf cohomology, which will be discussed below in this note.

We recall a little discussions from homological algebra (or algebraic topology) that will be frequently used later in this note.

Recall: Let $\mathbf{C}_\bullet = (C_n, d_n)$ and $\mathbf{D}_\bullet = (D_n, d'_n)$ be two complexes of R -modules, then a chain map between $\mathbf{C}_\bullet$ and $\mathbf{D}_\bullet$ is a collection of R -module homomorphisms $f = (f_n : C_n \rightarrow D_n)$ such that the

following diagram is commutative

$$\begin{array}{ccccccc}
 \dots & \longrightarrow & C_{n+1} & \xrightarrow{d_{n+1}} & C_n & \xrightarrow{d_n} & C_{n-1} \xrightarrow{d_{n-1}} \dots \\
 & & \downarrow f_{n+1} & & \downarrow f_n & & \downarrow f_{n-1} \\
 \dots & \longrightarrow & D_{n+1} & \xrightarrow{d'_{n+1}} & D_n & \xrightarrow{d'_n} & D_{n-1} \xrightarrow{d'_{n-1}} \dots
 \end{array}$$

then f_n maps cycles to cycles and boundaries to boundaries, i.e.,

$$f_n : Z_n(\mathbf{C}_\bullet) \longrightarrow Z_n(\mathbf{D}_\bullet), \quad f_n : B_n(\mathbf{C}_\bullet) \longrightarrow B_n(\mathbf{D}_\bullet)$$

hence $f : \mathbf{C}_\bullet \rightarrow \mathbf{D}_\bullet$ induces a homomorphism on the homological modules

$$f_* : H_n(\mathbf{C}_\bullet) \longrightarrow H_n(\mathbf{D}_\bullet)$$

$$[c] \mapsto [f_n(c)]$$

which is well-defined. Now, let $f, g : \mathbf{C}_\bullet \rightarrow \mathbf{D}_\bullet$ be two chain maps, if there is a collection of homomorphisms $s = (s_n : C_n \rightarrow D_{n+1})$ such that

$$f_n - g_n = d'_{n+1} \circ s_n + s_{n-1} \circ d_n \quad (\forall n)$$

then we say f and g are homotopic and denote $f \simeq g : \mathbf{C}_\bullet \rightarrow \mathbf{D}_\bullet$, then by the knowledge from homological algebra or algebraic topology we know that homotopic chain maps induce the same homomorphism on the homological modules, i.e.,

$$f \simeq g : \mathbf{C}_\bullet \rightarrow \mathbf{D}_\bullet \Rightarrow f_* = g_* : H_n(\mathbf{C}_\bullet) \rightarrow H_n(\mathbf{D}_\bullet)$$

It's a standard consequence from the knowledge of homological algebra or algebraic topology that the short exact sequence of complexes

$$0 \longrightarrow \mathbf{A}_\bullet \xrightarrow{f} \mathbf{B}_\bullet \xrightarrow{g} \mathbf{C}_\bullet \longrightarrow 0$$

i.e., the following diagram is commutative

$$\begin{array}{ccccccc}
 & & \vdots & & \vdots & & \vdots \\
 & & \downarrow d_{n+2}^A & & \downarrow d_{n+2}^B & & \downarrow d_{n+2}^C \\
 0 & \longrightarrow & A_{n+1} & \xrightarrow{f_{n+1}} & B_{n+1} & \xrightarrow{g_{n+1}} & C_{n+1} \longrightarrow 0 \\
 & & \downarrow d_{n+1}^A & & \downarrow d_{n+1}^B & & \downarrow d_{n+1}^C \\
 0 & \longrightarrow & A_n & \xrightarrow{f_n} & B_n & \xrightarrow{g_n} & C_n \longrightarrow 0 \\
 & & \downarrow d_n^A & & \downarrow d_n^B & & \downarrow d_n^C \\
 & & \vdots & & \vdots & & \vdots
 \end{array}$$

then it induces a long exact sequence of the homological modules

$$\dots \longrightarrow H_{n+1}(\mathbf{C}_\bullet) \xrightarrow{\delta_{n+1}} H_n(\mathbf{A}_\bullet) \xrightarrow{f_*} H_n(\mathbf{B}_\bullet) \xrightarrow{g_*} H_n(\mathbf{C}_\bullet) \xrightarrow{\delta_n} H_{n-1}(\mathbf{A}_\bullet) \longrightarrow \dots$$

where $\delta_n : H_n(\mathbf{C}_\bullet) \rightarrow H_{n-1}(\mathbf{A}_\bullet)$ is the connecting homomorphism. Moreover, if we have the following commutative diagram of complexes with exact rows

$$\begin{array}{ccccccc} 0 & \longrightarrow & \mathbf{A}_\bullet & \xrightarrow{i} & \mathbf{B}_\bullet & \xrightarrow{p} & \mathbf{C}_\bullet \longrightarrow 0 \\ & & \downarrow f & & \downarrow g & & \downarrow h \\ 0 & \longrightarrow & \mathbf{A}'_\bullet & \xrightarrow{j} & \mathbf{B}'_\bullet & \xrightarrow{q} & \mathbf{C}'_\bullet \longrightarrow 0 \end{array}$$

then we have the following commutative diagram with exact rows

$$\begin{array}{ccccccccccc} \dots & \longrightarrow & H_n(\mathbf{A}_\bullet) & \xrightarrow{i_*} & H_n(\mathbf{B}_\bullet) & \xrightarrow{p_*} & H_n(\mathbf{C}_\bullet) & \xrightarrow{\delta_n} & H_{n-1}(\mathbf{A}_\bullet) & \longrightarrow & \dots \\ & & \downarrow f_* & & \downarrow g_* & & \downarrow h_* & & \downarrow f_* & & \\ \dots & \longrightarrow & H_n(\mathbf{A}'_\bullet) & \xrightarrow{j_*} & H_n(\mathbf{B}'_\bullet) & \xrightarrow{q_*} & H_n(\mathbf{C}'_\bullet) & \xrightarrow{\delta'_n} & H_{n-1}(\mathbf{A}'_\bullet) & \longrightarrow & \dots \end{array}$$

Definition 17. Let $\mathcal{U} = \{U_\alpha\}_{\alpha \in A}$ be an indexed open cover for the topological space X and $\mathcal{V} = \{V_\beta\}_{\beta \in B}$ is another indexed open cover for X . If for any $\alpha \in A$, there is a $\beta \in B$ such that $V_\beta \subseteq U_\alpha$, then we say $\mathcal{V}$ is a **refinement** of $\mathcal{U}$. Moreover, we define the **refining map**

$$\rho : B \longrightarrow A$$

such that $V_\beta \subseteq U_{\rho(\beta)}$ for all $\beta \in B$.

Remark: Let $\mathcal{U}, \mathcal{V}, \rho$ be as in the definition above, let $\mathcal{S}$ be a sheaf on X , then we define the map

$$\begin{aligned} \rho^\# : C^p(\mathcal{U}; \mathcal{S}) &\longrightarrow C^p(\mathcal{V}; \mathcal{S}) \\ c &\mapsto \rho^\# c \end{aligned}$$

where

$$(\rho^\# c)_{\beta_0 \dots \beta_p} := c_{\rho(\beta_0) \dots \rho(\beta_p)}|_{V_{\beta_0} \cap \dots \cap V_{\beta_p}}$$

for any multi-index $(\beta_0, \beta_1, \dots, \beta_p)$ of length $p + 1$.

Proposition 2. The map $\rho^\# : C^p(\mathcal{U}; \mathcal{S}) \longrightarrow C^p(\mathcal{V}; \mathcal{S})$ defined above is commutative with the coboundary operator δ , i.e.,

$$\rho^\# \circ \delta = \delta \circ \rho^\#$$

Proof: Straightforward verification by the definition of δ and $\rho^\#$. $\square$

Remark: This proposition tells us that we have the following commutative diagram

$$\begin{array}{ccccccc} \dots & \xrightarrow{\delta} & C^p(\mathcal{U}; \mathcal{S}) & \xrightarrow{\delta} & C^{p+1}(\mathcal{U}; \mathcal{S}) & \xrightarrow{\delta} & \dots \\ & & \downarrow \rho^\# & & \downarrow \rho^\# & & \\ \dots & \xrightarrow{\delta} & C^p(\mathcal{V}; \mathcal{S}) & \xrightarrow{\delta} & C^{p+1}(\mathcal{V}; \mathcal{S}) & \xrightarrow{\delta} & \dots \end{array}$$

so by our standard discussion above, $\rho^\#$ induces a homomorphism

$$\rho^* : H^p(\mathcal{U}; \mathcal{S}) \longrightarrow H^p(\mathcal{V}; \mathcal{S})$$

Actually, the homomorphism ρ^* induced by $\rho^\#$ is independent of the choice of the refining map as the following lemma says.

Lemma 2. Let $\mathcal{U}, \mathcal{V}, \rho$ be as above, then the homomorphism ρ^* is independent of the choice of the refining map $\rho : B \rightarrow A$.

Proof: (**sketch**) Let $\mathcal{U} = \{U_\alpha\}_{\alpha \in A}, \mathcal{V} = \{V_\beta\}_{\beta \in B}$ be as above and $\rho, \tilde{\rho} : B \rightarrow A$ be two different refining maps. Let $\mathcal{S}$ be a sheaf on X . For any $p \in \mathbb{Z}_+$, define the following homomorphism

$$\theta : C^p(\mathcal{U}; \mathcal{S}) \longrightarrow C^{p-1}(\mathcal{V}; \mathcal{S})$$

$$c \mapsto \theta c$$

where

$$(\theta c)_{\beta_0 \dots \beta_{p-1}} := \sum_{j=0}^{p-1} (-1)^j c_{\rho(\beta_0) \dots \rho(\beta_j) \tilde{\rho}(\beta_{j+1}) \dots \tilde{\rho}(\beta_{p-1})} |_{V_{\beta_0} \cap \dots \cap V_{\beta_{p-1}}}$$

then it can be shown that

$$\theta \circ \delta + \delta \circ \theta = \tilde{\rho}^\# - \rho^\#$$

which means that $\tilde{\rho}^\# \simeq \rho^\# : \mathbf{C}^\bullet(\mathcal{U}; \mathcal{S}) \rightarrow \mathbf{C}^\bullet(\mathcal{V}; \mathcal{S})$ and hence by our standard discussion above, we know that

$$\tilde{\rho}^* = \rho^* : H^p(\mathcal{U}; \mathcal{S}) \longrightarrow H^p(\mathcal{V}; \mathcal{S})$$

i.e., independent of the choice of the refining map. $\square$

Remark: This lemma tells us that for a refinement $\mathcal{V}$ of $\mathcal{U}$, then we have a canonical homomorphism, which we may denote by $\rho^* \triangleq \rho_{\mathcal{U}\mathcal{V}}^* : H^p(\mathcal{U}; \mathcal{S}) \rightarrow H^p(\mathcal{V}; \mathcal{S})$. If $\mathcal{W} = \{W_\gamma\}_{\gamma \in C}$ is a refinement of $\mathcal{V} = \{V_\beta\}_{\beta \in B}$ and $\mathcal{V} = \{V_\beta\}_{\beta \in B}$ is a refinement of $\mathcal{U} = \{U_\alpha\}_{\alpha \in A}$, then there are refining maps such that $\rho_{\mathcal{U}\mathcal{W}}^* = \rho_{\mathcal{V}\mathcal{W}}^* \circ \rho_{\mathcal{U}\mathcal{V}}^*$, so if we define

$$\mathcal{U} \leq \mathcal{V} \iff \mathcal{V} \text{ is a refinement of } \mathcal{U}$$

then the collection $\{H^p(\mathcal{U}; \mathcal{S}) \mid \mathcal{U} \text{ is an open cover of } X\}$ indexed by $\mathcal{U}$ together with the homomorphism $\rho_{\mathcal{U}\mathcal{V}}^* : H^p(\mathcal{U}; \mathcal{S}) \longrightarrow H^p(\mathcal{V}; \mathcal{S})$ forms a direct system, which yields the following definition.

Definition 18. Let $\mathcal{S}$ be a sheaf on the topological space X . The p -th sheaf cohomology group of X with coefficients in $\mathcal{S}$ (or Čech cohomology groups with coefficients in $\mathcal{S}$) is defined by

$$H^p(X; \mathcal{S}) := \varinjlim_{\mathcal{U}} H^p(\mathcal{U}; \mathcal{S})$$

Remark: For any $[c] \in H^p(\mathcal{U}; \mathcal{S})$, we denote the equivalence class of $[c]$ in $H^p(X; \mathcal{S})$ by $[[c]]$. By the definition, we know that if $c_1, c_2 \in Z^p(\mathcal{U}; \mathcal{S})$, then $[[c_1]] = [[c_2]] \in H^p(X; \mathcal{S}) \iff$ there is a refinement $\mathcal{V}$ of $\mathcal{U}$ such that $\rho_{\mathcal{U}\mathcal{V}}^*[c_1] = \rho_{\mathcal{U}\mathcal{V}}^*[c_2]$ (or equivalently, there is a refining map $\rho : B \rightarrow A$ such that $\rho^\#(c_1 - c_2) \in B^p(\mathcal{V}; \mathcal{S})$)

Let $\mathcal{S}, \mathcal{T}$ be two sheaves on the topological space X , let $F : \mathcal{S} \rightarrow \mathcal{T}$ be a sheaf morphism, for any open cover $\mathcal{U} = \{U_\alpha\}$ of X , we define

$$F_\# : C^p(\mathcal{U}; \mathcal{S}) \longrightarrow C^p(\mathcal{U}; \mathcal{T})$$

$$c \mapsto F_\# c$$

where

$$(F_\# c)_{\alpha_0 \dots \alpha_p} := F(c_{\alpha_0 \dots \alpha_p})$$

for any multi-index $(\alpha_0, \dots, \alpha_p)$ of length $p+1$. By the definition of δ and $F_\#$, it's a straightforward computation to show that

$$F_\# \circ \delta = \delta \circ F_\#$$

so by our previous standard discussion, it induces a homomorphism, which we denote as

$$\widehat{F}_\# : H^p(\mathcal{U}; \mathcal{S}) \longrightarrow H^p(\mathcal{U}; \mathcal{T})$$

i.e., $\widehat{F}_\#([c]) = [F_\#(c)]$. If $\mathcal{V}$ is a refinement of $\mathcal{U}$, then we can choose a refining map $\rho : B \rightarrow A$, again by the definition of ρ^* and $\widehat{F}_\#$, it can be shown that

$$\rho^* \circ \widehat{F}_\# = \widehat{F}_\# \circ \rho^*$$

i.e., the diagram is commutative

$$\begin{array}{ccccccc} \dots & \longrightarrow & H^p(\mathcal{U}; \mathcal{S}) & \xrightarrow{\rho_{\mathcal{U}\mathcal{V}}^*} & H^p(\mathcal{V}; \mathcal{S}) & \longrightarrow & \dots \\ & & \downarrow \widehat{F}_\# & & \downarrow \widehat{F}_\# & & \\ \dots & \longrightarrow & H^p(\mathcal{U}; \mathcal{T}) & \xrightarrow{\rho_{\mathcal{U}\mathcal{V}}^*} & H^p(\mathcal{V}; \mathcal{T}) & \longrightarrow & \dots \end{array}$$

since $H^p(X; \mathcal{S}) := \varinjlim_{\mathcal{U}} H^p(\mathcal{U}; \mathcal{S})$, so $\widehat{F}_\#$ induces a homomorphism

$$F_* : H^p(X; \mathcal{S}) \longrightarrow H^p(X; \mathcal{T})$$

moreover, the following result is easy to see.

Theorem 2. Let $\mathcal{S}, \mathcal{T}$ be two sheaves on the topological space X , let $F : \mathcal{S} \rightarrow \mathcal{T}$ be a sheaf morphism. The for any $p \in \mathbb{Z}_+$, the induced homomorphism $F_* : H^p(X; \mathcal{S}) \rightarrow H^p(X; \mathcal{T})$ discussed above satisfies

$$(\text{Id}_{\mathcal{S}})_* = \text{Id}_{H^p(X; \mathcal{S})}, \quad (F \circ G)_* = F_* \circ G_*$$

Remark: This theorem tells us that the functor $H^p(X; \cdot)$ is a contravariant functor from the category $\mathbf{Sh}(X)$ of sheaves on X to the category of abelian groups $\mathbf{Ab}$, where $H^p(X; \cdot)(\mathcal{S}) := H^p(X; \mathcal{S})$ and $H^p(X; \cdot)(F) := F_*$.

In the case of degree 0, we have the following result.

Proposition 3. Let $\mathcal{S}$ be a sheaf on the topological space X , then there is an isomorphism

$$I_{\mathcal{S}} : \mathcal{S}(X) \xrightarrow{\cong} H^0(X; \mathcal{S})$$

such that for any sheaf morphism $F : \mathcal{S} \rightarrow \mathcal{T}$, we have the following commutative diagram

$$\begin{array}{ccc} \mathcal{S}(X) & \xrightarrow[\cong]{I_{\mathcal{S}}} & H^0(X; \mathcal{S}) \\ F_X \downarrow & & \downarrow F_* \\ \mathcal{T}(X) & \xrightarrow[\cong]{I_{\mathcal{T}}} & H^0(X; \mathcal{T}) \end{array}$$

Proof: By Proposition 1, we know that

$$I_{\mathcal{U}, \mathcal{S}} : \mathcal{S}(X) \longrightarrow H^0(\mathcal{U}; \mathcal{S})$$

is an isomorphism, then by the definition of ρ^* and $I_{\mathcal{U}, \mathcal{S}}$, we can verify that

$$\rho_{\mathcal{U}, \mathcal{V}}^* \circ I_{\mathcal{U}, \mathcal{S}} = I_{\mathcal{V}, \mathcal{S}}$$

i.e., the diagram commutes

$$\begin{array}{ccc} H^0(\mathcal{U}; \mathcal{S}) & \xrightarrow{\rho_{\mathcal{U}, \mathcal{V}}^*} & H^0(\mathcal{V}; \mathcal{S}) \\ \cong \uparrow I_{\mathcal{U}, \mathcal{S}} & \nearrow I_{\mathcal{V}, \mathcal{S}} & \\ \mathcal{S}(X) & & \end{array}$$

since $H^0(X; \mathcal{S}) := \varinjlim_{\mathcal{U}} H^0(\mathcal{U}; \mathcal{S})$, so by taking direct limit, we get an isomorphism, which we denote as

$$I_{\mathcal{S}} : \mathcal{S}(X) \xrightarrow{\cong} H^0(X; \mathcal{S})$$

Now for a sheaf morphism $F : \mathcal{S} \rightarrow \mathcal{T}$, by our previous definition, we know that we have the map $\widehat{F}_{\#} : H^0(\mathcal{U}; \mathcal{S}) = Z^0(\mathcal{U}; \mathcal{S}) \rightarrow H^0(\mathcal{U}; \mathcal{T}) = Z^0(\mathcal{U}; \mathcal{T})$ which is given by $(\widehat{F}_{\#}(c))_{\alpha} = F(c_{\alpha})$, then it's straightforward verification that we have $\widehat{F}_{\#} \circ I_{\mathcal{U}, \mathcal{S}} = I_{\mathcal{U}, \mathcal{T}} \circ F_X$, i.e., the diagram is commutative

$$\begin{array}{ccc} \mathcal{S}(X) & \xrightarrow[\cong]{I_{\mathcal{U}, \mathcal{S}}} & H^0(\mathcal{U}; \mathcal{S}) \\ F_X \downarrow & & \downarrow \widehat{F}_{\#} \\ \mathcal{T}(X) & \xrightarrow[\cong]{I_{\mathcal{U}, \mathcal{T}}} & H^0(\mathcal{U}; \mathcal{T}) \end{array}$$

pass to the direct limit, we get

$$\begin{array}{ccc} \mathcal{S}(X) & \xrightarrow[\cong]{I_{\mathcal{S}}} & H^0(X; \mathcal{S}) \\ F_X \downarrow & & \downarrow F_* \\ \mathcal{T}(X) & \xrightarrow[\cong]{I_{\mathcal{T}}} & H^0(X; \mathcal{T}) \end{array}$$

so the proposition holds. $\square$

There is an analogue to our previous standard discussion from homological algebra (or algebraic topology), which is stated as below. The idea of the proof is similar to that in homological algebra (or algebraic topology).

Theorem 3. Let

$$0 \longrightarrow \mathcal{A} \xrightarrow{F} \mathcal{B} \xrightarrow{G} \mathcal{C} \longrightarrow 0$$

be a short exact sequence of sheaves on a paracompact Hausdorff space X , then we have the following long exact sequence

$$\begin{aligned} 0 \longrightarrow H^0(X; \mathcal{A}) \xrightarrow{F_*} H^0(X; \mathcal{B}) \xrightarrow{G_*} H^0(X; \mathcal{C}) \xrightarrow{\delta_*} H^1(X; \mathcal{A}) \xrightarrow{F_*} \dots \\ \dots \xrightarrow{F_*} H^p(X; \mathcal{B}) \xrightarrow{G_*} H^p(X; \mathcal{C}) \xrightarrow{\delta_*} H^{p+1}(X; \mathcal{A}) \xrightarrow{F_*} \dots \end{aligned}$$

where δ_* is the connecting homomorphism.

Theorem 4. Suppose we have the following commutative diagram of sheaves on the paracompact Hausdorff space X ,

$$\begin{array}{ccccccc} 0 & \longrightarrow & \mathcal{A} & \xrightarrow{F} & \mathcal{B} & \xrightarrow{g} & \mathcal{C} \longrightarrow 0 \\ & & \downarrow f & & \downarrow B & & \downarrow h \\ 0 & \longrightarrow & \mathcal{A}' & \xrightarrow{L} & \mathcal{B}' & \xrightarrow{T} & \mathcal{C}' \longrightarrow 0 \end{array}$$

where the exact rows are exact, then we have the following commutative diagram concerning the connecting homomorphisms

$$\begin{array}{ccc} H^p(X; \mathcal{C}) & \xrightarrow{\delta_*} & H^{p+1}(X; \mathcal{A}) \\ \downarrow h_* & & \downarrow f_* \\ H^p(X; \mathcal{C}') & \xrightarrow{\delta_*} & H^{p+1}(X; \mathcal{A}') \end{array}$$

1.7 Acyclic Resolution

Definition 19. Let $\mathcal{S}$ be a sheaf on the topological space X . A **resolution of $\mathcal{S}$** is an exact sequence of sheaves on X of the form

$$0 \longrightarrow \mathcal{S} \longrightarrow \mathcal{A}^0 \longrightarrow \mathcal{A}^1 \longrightarrow \mathcal{A}^2 \longrightarrow \dots$$

Definition 20. A sheaf $\mathcal{S}$ on the topological space X is called **acyclic** if

$$H^p(X; \mathcal{S}) = 0 \quad \text{for } \forall p \geq 1$$

Definition 21. A resolution of the sheaf $\mathcal{S}$ on X is called an **acyclic resolution** if each $\mathcal{A}^k$ is acyclic, i.e.,

$$H^p(X; \mathcal{A}^k) = 0 \quad (\forall p \geq 1, \forall k \geq 0)$$

Latexed by 华夏文化